Digital Egypt Pioneers

Project Proposal: Rafeeq

Browser-Based Accessibility Extension & Website

Rafeeq is an adaptive, multi-modal digital ecosystem that bridges communication gaps for people with diverse disabilities: Deaf/Mute, Blind, Dyslexic, and Aphasic users. Rather than a standalone mobile app, Rafeeq is architected as a **lightweight browser extension** that injects assistive UI overlays and real-time processing directly into mainstream web communication platforms such as WhatsApp Web and Facebook meeting users where they already communicate.

A. Computer Vision Track — Visual Sensors & Intake

Real-time video/image processing and local frame-to-token extraction via Edge AI.

Deaf & Mute Support

- Tracks hand, arm & facial landmarks (MediaPipe Holistic / OpenPose) for continuous sign language translation.
- Continuous sign boundary detection + lip reading via 3D CNNs.

Blind Support

- Object detection (YOLOv8), segmentation & depth estimation for environmental awareness and hazard tracking.
- Spatial audio conversion, facial emotion recognition, and Visual Question Answering (VQA).
- Currency recognition to identify banknotes and coins.

Dyslexia Support

- OpenCV preprocessing (binarization, de-skewing, noise removal) + OCR (EasyOCR/Tesseract).
- LayoutLM for document structure; re-renders clean text in accessible typefaces (OpenDyslexic).

Aphasia Support

- Tracks visual UI states and touch patterns to drive predictive quick-reply categorization and fast symbol/text output.

Cross-Disability Bridge

- Ingests sign-language camera streams and translates them via low-latency WebSockets into text or voice.

B. LLM & RAG Track — Text Normalization & Retrieval

Handles language refinement and domain-specific lookups downstream of the CV track.

Text Normalization (LLM)

- Lightweight open-weight models (Qwen3 4B/8B) act strictly as a semantic normalizer.
- Principle: “Improve the form, preserve the information” — cleans noisy CV output without inventing facts or changing intent.

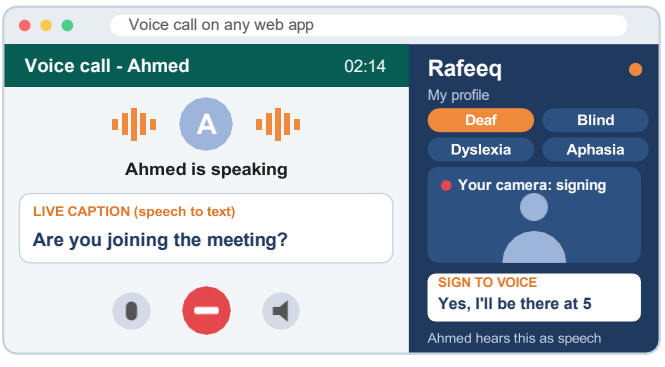
Contextual Retrieval (RAG) (Optional)

- Dedicated knowledge-access layer.
- Pulls verified domain info — accessibility guidelines, platform help manuals, sign dialects — to ground responses.

Integration & Deployment Architecture

- Operates as a browser extension that reads DOM elements or video streams from target web apps.
- Dynamically modifies the interface based on the user's selected accessibility profile.
- Processes inputs locally (Edge AI) or via secure streaming middleware for low-latency, privacy-respecting performance.
- A companion website (accessibility center, unified inbox, profiles) manages user settings.

Rafeeq in Action: Voice Call (Extension)



Rafeeq Landing Page (Website)

